Q14. Autopsy metadata, deleted files and keyword analysis

Original question: Analyze a forensic disk image using Autopsy. Examine file metadata and investigate available deleted files, keyword matches, or other relevant artifacts. Document the evidence identified.

Aim

Inspect metadata, available deleted entries and keyword matches in a supplied image, and document relevant artifacts without promising recovery.

Requirements

Supplied image and hash, Autopsy case storage and scenario-relevant keywords. Use the Autopsy edition distinction in Q8. References: [R5]-[R7].

Procedure

Step 1 - Prepare an independent case record

```
mkdir -p "$HOME/EHF_Exam/Q14"
cd "$HOME/EHF_Exam/Q14"
IMAGE='<DISK_IMAGE_PATH>'
sha256sum "$IMAGE" | tee image_before.sha256
```

Open an approved existing case for this image or create EHF_Q14 using Q8's case/add-image steps. Confirm the selected data source, timezone and completed ingest. Never assume the selected GUI case corresponds to the file you hashed.

Step 2 - Examine file metadata

1. Expand Data Sources, choose a volume and browse to a relevant file.
2. Select the file and open File Metadata/Properties in the content viewer. Record evidence path, metadata address, size, allocation/deletion status, file type and available created/modified/accessed/changed timestamps.
3. Inspect Text or Hex content without opening the file in its native application. Compare extension with detected type when available. A mismatch is a lead, not proof of malicious content.
4. Note timezone, timestamp precision and clock assumptions. Unix ctime normally records metadata change, not creation; birth time exists only on some filesystems. Windows and Unix timestamp semantics differ.

Step 3 - Investigate deleted entries

1. In modern Autopsy, look under Views > Deleted Files, or the available deleted-file grouping. Confirm the relevant volume. In legacy Autopsy use File Analysis and deleted-entry views; names and navigation vary.
2. Select an actual deleted entry. Record its original path/name if surviving, metadata identifier, status and available size/times. Preview its content.
3. If supported, right-click and choose Extract File(s) or the edition's export action. Save outside evidence to the Q14 output directory. Record read errors or partial content. Do not execute the export.
4. A deleted name can survive while content is overwritten, trimmed, fragmented or inaccessible. If preview/export fails, document that outcome instead of calling it a recovered file.

Step 4 - Search relevant keywords

1. In modern Autopsy use Keyword Search, create/select a small list of examiner-relevant terms, choose literal versus regex matching deliberately and the correct data source.
2. Run the search or required keyword ingest/indexing. Wait for completion; examine Keyword Hits and the surrounding text and source file. Index coverage depends on file types and extraction success.
3. In legacy Autopsy use its Keyword Search interface; it does not share modern indexing controls. Record search mode/encoding and scope.
4. Bookmark/tag supported hits, retaining keyword, source path, identifier and hit context. A search hit alone does not establish intent or authorship.

Step 5 - Hash exports and verify the image

```
# Run only after setting this to an actual exported file:
EXPORT='<EXPORTED_FILE_PATH>'
sha256sum "$EXPORT" | tee export_hash.txt
sha256sum -c image_before.sha256
```

Skip the export-hash command if no file was exported; explicitly record that no recoverable content was obtained.

Expected output - illustrative worksheet

```
M1: [path / metadata address / size / timestamps + timezone]
D1: [deleted entry / preview status / exported bytes if any]
K1: [keyword / exact context / source / relevance]
```

No metadata, deleted entry or keyword match has been obtained from an input in this task. Populate only supported rows; an absent category is valid.

Interpretation, observation and result

Distinguish an existing file, a deleted metadata entry, recovered content and a search hit. Correlate time, location and content before drawing a conclusion. Explain recovery completeness and uncertainty. The result is a documented metadata/deletion/search examination, not a guarantee that every deleted file can be recovered.

Common errors

- No deleted view: check edition, volume and allocation filters; there may be no surviving entries.
- No hits: verify keyword spelling, encoding, indexing status, file extraction and scope.
- Export is unreadable: content may be partial, overwritten, encrypted or inherently binary; do not claim intact recovery.

Viva questions

- Does deleted mean erased? No; metadata/content may survive until reused, but recovery is uncertain.
- Is ctime always creation time? No; on Unix it normally means inode status-change time.
- What makes a keyword hit meaningful? Relevant source context and corroboration.
- Why hash an export? To identify and verify that derivative file in later analysis.